RANDOM VARIABLES PRACTICAL

Exercise 1 [⭐⭐] Using R Studio or otherwise, calculate the following quantities.

a) $P(|X| < 2)$ for $X \sim N(1,4)$ **[0.6246553]**

b) $P(X^2 > 10)$ for $X \sim N(3,0.4)$ **[0.3987502]**

c) q such that $P(|X-1| \geq q) = 0.2$ for $X \sim N(1,3)$ **[2.219712]**

d) q such that $P((X+2)^2 \geq q) = 0.5$ for $X \sim N(-2,9)$ **[4.094428]**

e) q such that $P(|X| < q) = 0.80$ for $X \sim N(20,2)$ **[21.190230]**

Exercise 2 [⭐⭐] Using R Studio or otherwise, calculate the following quantities.

a) $P(X < 3)$ for $X \sim Exp(1)$ **[0.9502129]**

b) $P(1 < X^2 < 3)$ for $X \sim Exp(0.5)$ **[0.1859106]**

c) $P(|X| > 4)$ for $X \sim Exp(0.2)$ **[0.449329]**

d) q such that $P(X \geq q) = 0.05$ for $X \sim Exp(0.2)$, **[14.97866]**

e) q such that $P(\exp(X+1) \leq q) = 0.85$ for $X \sim Exp(2)$ **[7.018574]**

Exercise 3 [⭐⭐] Using R Studio or otherwise, calculate the following quantities.

a) $P(X < 3)$ for $X \sim$ Uni$(1,4)$ **[2/3]**

b) $P(1 < X^2 < 3)$ for $X \sim$ Uni$(-2,6)$ **[0.1830128]**

c) $P(X^2 > 4)$ for $X \sim$ Uni$(-4,6)$ **[6/10]**

Exercise 4 [⭐⭐] Using R Studio or otherwise, calculate the following quantities.

a) $P(X < 0.85)$ for $X \sim$ Gamma$(1,2)$ **[0.8173165]**

b) $P(|X| > 4)$ for $X \sim$ Gamma$(5,1)$ **[0.6288369]**

c) q such that $P(X \geq q) = 0.05$ for $X \sim$ Gamma$(5,1)$ **[9.153519]**

d) q such that $P(\sqrt{X+1} \geq q) = 0.85$ for $X \sim$ Gamma$(2,3)$ **[1.108037]**

e) Using R Studio, produce an (i.i.d.) sample $x_1, x_2, \ldots, x_n$ of size $n = 10{,}000$ such that $x_i \sim$ Gamma$(1,3)$. What is the sample mean of your sample? **[Should be around 0.33]**